

Learning Human-like Finger Gaiting on an Anthropomorphic Hand

Kairui Yang, Dongjie Jiang, Lecheng Ruan[✉], and Qining Wang[✉]

School of Advanced Manufacturing and Robotics, Peking University, Beijing 100871, China

✉ ruanlecheng@ucla.edu, qiningwang@pku.edu.cn

Abstract. A key challenge in dexterous non-prehensile manipulation lies in dynamic finger gaiting—the sequential repositioning of fingers to achieve continuous object motion. The pen-spinning task, requiring precise, sequential multi-finger coordination without a stable grasp, serves as an ideal testbed for investigating such gaiting. Prior work, often limited by hand morphology, has typically yielded simpler policies reliant on fingertip balancing rather than dynamic finger gaiting. In this work, we investigate learning finger gaiting on an anthropomorphic hand in simulation. However, achieving this skill through reinforcement learning (RL) introduces significant challenges, particularly in policy exploration and the processing of complex observations. To address these, our framework employs waypoint-guided initialization and utilizes normalized contact forces as a form of privileged information during training. Our simulation results demonstrate the emergence of dynamic finger gaiting, enabling the efficient execution of the pen-spinning task. This work thereby establishes a viable methodology for acquiring complex coordination skills on high-degree-of-freedom (DoF) anthropomorphic hands.

Keywords: Finger Gaiting, Dexterous Manipulation, Non-prehensile Manipulation, Reinforcement Learning

1 Introduction

The pursuit of human-level dexterity is a primary driver of advancements in robotic manipulation. While robots excel in prehensile tasks involving firm grasps [1][2][3], they are often challenged by non-prehensile manipulation. These tasks, such as reorienting, rolling, or spinning objects, demand dynamic control without a secure grasp [4][5]. Finger gaiting, a central focus of this paper, is a key strategy for such non-prehensile scenarios. This bio-inspired approach involves coordinated, sequential finger contacts to maintain continuous object control through alternating support and propulsion phases [6]. This technique enables object manipulation beyond a fixed grasp, which is fundamental for expanding robotic dexterity beyond static paradigms and for achieving versatile interaction in unstructured environments.

We investigate the complexities of finger gaiting using pen-spinning as a representative task [7]. This task is particularly suitable because it demands

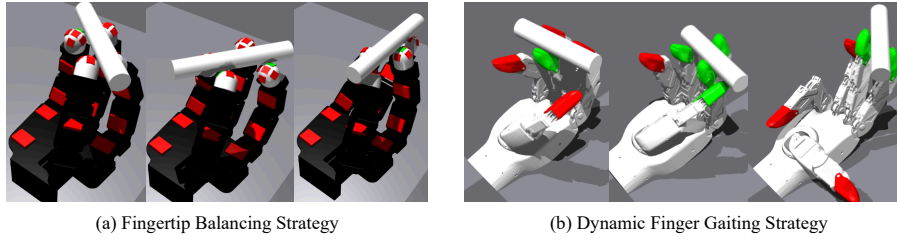


Fig. 1: Influence of hand morphology on pen-spinning. (a) A hand with limited DoF and broad fingertips defaults to a simpler fingertip balancing strategy. (b) In contrast, a high-DoF anthropomorphic hand enables a more complex and human-like dynamic finger gaiting strategy.

precise temporal coordination of multiple fingers and continuous, nuanced force modulation—core challenges in dynamic non-prehensile manipulation.

The ability to execute such intricate maneuvers is highly dependent on hand morphology. Prior work using less anthropomorphic hands (Fig. 1(a)) often yielded simpler policies restricted to fingertip balancing [7]. In this work, we employ a high-DoF anthropomorphic hand (Fig. 1(b)), observing that its superior dexterity and versatile contact surfaces (e.g., finger pads and tips) are essential for enabling the development of more sophisticated, human-like manipulation strategies.

However, this morphological complexity also introduces significant algorithmic challenges for learning. The vast action space of a high-DoF system complicates policy learning, a difficulty exacerbated in non-prehensile tasks that lack the inherent stability of a firm grasp. Although RL is emerging as a powerful paradigm for managing high-dimensional action spaces in dexterous manipulation [8][9], learning sophisticated finger gaiting on anthropomorphic hands remains hindered by the following critical challenges:

1. **Ineffective Exploration in High-Dimensional Spaces:** The vast action space of a high-DoF hand renders standard exploration methods, such as random joint sampling, ineffective. For a task like finger gaiting, which requires specific initial contacts to begin, the probability of discovering a viable starting state through random exploration is exceedingly low. This necessitates a more guided approach to steer exploration toward productive regions of the state-action space.
2. **Difficulty Interpreting Complex Sensorimotor Information:** An anthropomorphic hand generates high-dimensional streams of proprioceptive and contact data. For dynamic control, the policy must learn to interpret this complex information and discern subtle yet critical differences—such as the force vectors required for stable support versus active propulsion. Processing this noisy, high-dimensional data to produce precise motor commands in real-time poses a substantial learning challenge.

To address these challenges, our framework integrates targeted solutions. We address the exploration challenge with a reference waypoint-guided initializa-

tion strategy, which leverages human demonstrations to bootstrap the learning process from viable initial states. To manage the complexity of sensorimotor control, we utilize privileged 3D net contact force information. This data is carefully processed and normalized to provide a clean, informative signal for the policy.

Our primary contributions are as follows:

1. **A Novel Learning Framework for Dynamic Finger Gaiting:** We present a complete RL-based framework that successfully combines waypoint-guided initialization and the strategic use of privileged contact information, enabling a high-DoF anthropomorphic hand to learn sophisticated finger gaiting.
2. **A Reference Waypoint-Guided Initialization Strategy:** We propose and validate an initialization strategy that uses reference waypoints derived from human demonstration. These waypoints represent critical contact transitions in the pen-spinning cycle, effectively bootstrapping the learning process for complex finger gaiting.
3. **Empirical Demonstration of Emergent Finger Gaiting:** We empirically demonstrate that our framework enables an anthropomorphic hand to learn sophisticated pen-spinning through emergent dynamic finger gaiting, achieved in just 1.5 hours of training. This result validates our approach and showcases its effectiveness in acquiring complex, human-like manipulation skills.

The remainder of this paper is structured as follows. Section 2 reviews related work on dexterous manipulation and finger gaiting. Section 3 details our proposed methodology, including the learning framework and reward design. Section 4 presents our experimental setup and results, followed by Section 5, which discusses the findings and outlines future research directions.

2 Related Work

2.1 Reinforcement Learning for In-hand Manipulation

Reinforcement learning has emerged as a foundational approach for unlocking complex skills in high-degree-of-freedom (DoF) dexterous hands [1]. Seminal research demonstrated that RL could achieve in-hand object reorientation in simulation with successful transfer to the real world, establishing a powerful paradigm for the field [2]. Building on this work, the research community has extended the complexity by tackling challenges such as bimanual manipulation [8], long-horizon tasks [10], and generalized reorientation skills [11]. This trend toward leveraging higher-DoF systems and addressing more intricate tasks underscores the ambition to replicate human-level dexterity.

However, this ambition has revealed persistent challenges. As the dimensionality of robotic hands increases, so does the complexity of exploration and control, making skill acquisition significantly more difficult [12]. Moreover, while these studies have achieved impressive functional outcomes, the learned behaviors do not typically exhibit the dynamic and nuanced strategies characteristic

of human manipulation, such as finger gaiting. Much of the existing work on in-hand manipulation [13][11][14] has focused on achieving stable control rather than eliciting the sophisticated, sequential finger movements that are central to our work.

2.2 Finger Gaiting as a Benchmark for Human-like Manipulation

Finger gaiting—the sequential repositioning of fingers to maintain continuous control of an object—is a hallmark of human dexterity, enabling manipulation beyond a static grasp [6]. This non-prehensile skill is fundamental for advanced interaction, and pen-spinning serves as an ideal benchmark task due to its stringent requirements for precise timing and dynamic force modulation [7].

Critically, the ability to develop such sophisticated strategies is deeply intertwined with the hand’s physical morphology. As observed in prior work [7], hand designs with limited DoF or broad, homogeneous fingertips often lead to simpler policies, such as fingertip balancing, which prioritize stability over agility (Fig. 1(a)). In contrast, dynamic finger gaiting involves a complex interplay of supporting and propulsive actions from different finger segments. This more dynamic, human-like behavior is hypothesized to emerge more readily on highly anthropomorphic hands that feature slender digits and the high DoF necessary for fine-grained, independent control (Fig. 1(b)). While the field is trending toward more capable hardware, learning to exploit these anthropomorphic features for dynamic gaiting remains a significant, unaddressed challenge. This paper specifically focuses on bridging this gap by investigating whether targeted RL strategies can elicit emergent finger gaiting on a high-DoF anthropomorphic hand.

3 Methodology

Our methodology builds upon foundational work that achieved pen-spinning on a robotic hand with fewer degrees of freedom [7]. We adapt and extend this approach to a high-DoF anthropomorphic hand, which presents a greater challenge in learning dynamic finger gaiting. Our approach centers on an RL framework enhanced by two key strategies designed to make this complex skill tractable: a waypoint-guided initialization strategy to bootstrap learning and the strategic use of privileged 3D net force information to master nuanced physical interactions.

3.1 Reinforcement Learning Framework

Settings and Notations We formulate the pen-spinning task as a standard reinforcement learning problem. The goal is to train a policy $\pi(\mathbf{a}_t|\mathbf{o}_t)$ that maps observations to actions to maximize a cumulative reward. We employ Proximal Policy Optimization (PPO) [15] as the core learning algorithm. The agent controls an anthropomorphic hand with N_{DoF} degrees of freedom.

The key notations are defined as follows:

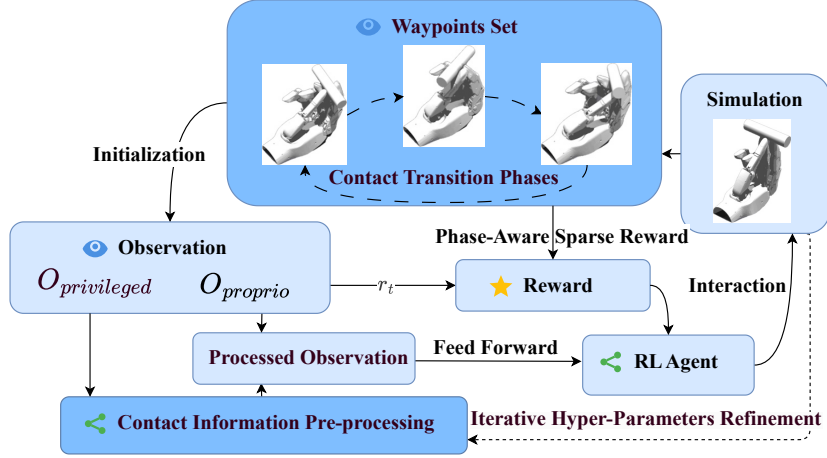


Fig. 2: The training pipeline for learning finger gaiting. The process begins with waypoint-guided initialization derived from a human demonstration. An RL agent then learns by interacting with the simulation, guided by proprioceptive and privileged observations. A key feature is the feedback loop that iteratively refines contact force normalization parameters based on interaction data to improve policy performance.

- **Observation (o_t):** The observation for the policy is composed of proprioceptive information O_{pro} (joint angles $\mathbf{q}$, velocities $\dot{\mathbf{q}}$, and previous target joint angles) and privileged information O_{pri} (fingertip positions, 3D net contact forces, object pose and velocities, and an object point cloud).
- **Action ($\mathbf{a}_t$):** The action space $\mathbf{a}_t = \Delta \mathbf{q}_{tgt} \in \mathbb{R}^{N_{DoF}}$ defines the incremental adjustments to the target joint angles. These adjustments are converted into joint torques $\boldsymbol{\tau}$ using a Proportional-Derivative (PD) controller, which then drive the simulation.

Reward Function The total reward is calculated using the following formula with empirically determined weights:

$$R_{tot} = w_{rot}r_{rot} + w_{sta}r_{sta} + w_{smo}r_{smo} + w_{vel}r_{vel} + w_{way}r_{way}$$

where r_{rot} encourages continuous rotation, r_{sta} penalizes deviations from the target height and orientation, r_{smo} promotes smooth movements, r_{vel} discourages excessive speed, and r_{way} is a sparse reward for reaching key manipulation stages. The terms w_{rot} , w_{sta} , w_{smo} , w_{vel} , and w_{way} represent the corresponding weights for each reward component.

3.2 Waypoint Guidance in Reinforcement Learning

Standard RL is often ineffective for complex, sequential tasks like finger gaiting due to the vast state-action space, which makes meaningful exploration highly improbable. This challenge is well-documented, and solutions often involve incorporating demonstrations or guided exploration [9]. To address this, we propose a *Waypoint-Guided Reinforcement Learning* strategy, which leverages human demonstration data in two ways: guiding exploration through targeted initialization and shaping the policy via sparse rewards.

First, to solve the exploration problem, the strategy provides effective initial states. We define reference waypoints by extracting them from a human pen-spinning trajectory. These waypoints represent crucial *transitional states* within the gaiting cycle, not merely static poses. These potential waypoints are refined by applying perturbations and evaluating the resulting configurations against criteria such as contact stability and robustness. Configurations scoring above a threshold are selected as initial waypoints for training episodes, which begin from states sampled from Gaussian distributions centered around them. This significantly biases exploration towards dynamically relevant regions of the state space.

Second, these same waypoints are used to provide a sparse *waypoint achievement reward* (r_{way}). This reward signal acts as an intermediate target, explicitly encouraging the policy to progress through the key transitional stages of the manipulation. By combining targeted initialization with reward-based guidance, this strategy accelerates learning and facilitates credit assignment. Such guidance is particularly critical for a high-DoF hand with slender fingertips, as it steers learning towards intricate finger gaiting rather than simpler balancing behaviors that might otherwise emerge [7].

3.3 Privileged Contact Information Pre-processing

Finger gaiting requires the precise modulation of contact forces to differentiate between supportive, propulsive, and guiding touches—a distinction critical for manipulating an object through sequential contacts. Binary contact information alone is inadequate for learning these nuanced interactions, as success depends on both the magnitude and direction of applied forces. Consequently, a hand model with heterogeneous fingertips capable of rich surface interactions benefits significantly from detailed net force data to master complex finger gaiting. In contrast, for hands with wider, more homogeneous fingertips that favor stable balancing over fine propulsion, simpler strategies reliant on fingertip balancing can emerge [7]. For these balancing approaches, binary contact data (indicating only the presence or absence of contact) can suffice, as they do not require the detailed force vector information essential for the propulsive and re-grasping actions of finger gaiting.

We incorporate detailed 3D net contact force vectors as part of the privileged information $\mathbf{O}_{\text{pri}}$. For each of the five fingertips, we consider the 3D net force

vector $\mathbf{f}_{\text{tip},i} = (F_x, F_y, F_z)_i \in \mathbb{R}^3$. These vectors are directly available in the simulation environment, resulting in a net contact force component within $\mathbf{O}_{\text{pri}}$.

Net Force Vector Normalization: Raw 3D net force values can vary widely in scale, potentially destabilizing the learning process. We apply a normalization scheme to each force component F_c (where $c \in \{x, y, z\}$ for each fingertip):

$$F'_c = \frac{\text{clip}(F_c, F_{\min}, F_{\max}) - F_{\min}}{F_{\max} - F_{\min}} \quad (1)$$

This scales the clipped force component to the range $[0, 1]$, where $F_{\min}$ and $F_{\max}$ are pre-set limits based on expected force ranges. An alternative mapping to the range $[-1, 1]$ using hyperbolic tangent normalization can also be employed:

$$F_{\text{norm},i} = \tanh(k \cdot F_i) \quad (2)$$

where the scaling factor k is empirically selected. Both techniques ensure that net force inputs are consistently scaled, helping the policy network interpret these critical signals for effective finger gaiting.

4 Experiments and Results

All experiments were performed within the Isaac Gym simulation environment on an NVIDIA RTX 4090 GPU. The simulation operated with a physics timestep of 5 ms and a control frequency of 20 Hz. The dexterous hand used in these experiments is the Linker Hand, a five-fingered hand with 21 DoF (Fig.1(b)). Its design is notably anthropomorphic, featuring slender fingertips and human-like pulp surfaces. The manipulated object was a cylindrical pen with a radius of 12 mm, a length of 120 mm, and a mass of 60 g.

We assessed multiple training configurations by varying initialization strategies and the method of processing contact information. As illustrated in Figure 3, we defined two primary initialization approaches: one using three waypoints from a human demonstration to guide dynamic learning, and another using six static balancing poses. For each initialization, we compared three methods for processing contact information: a baseline without force normalization, a method with a single, fixed normalization, and a strategy that iteratively refines the contact force normalization based on performance during training. We also tuned various learning hyperparameters, such as reward weights, and selected the optimal set for our final training strategy. The performance of these key configurations is detailed in Figure 3(c).

4.1 Validation of Reference Waypoint-Guided Initialization

Strategic initialization using reference waypoints derived from human demonstration proved critical to successful policy learning.

Comparison with Random Initialization: Ablation studies confirmed that policies trained with random initialization, i.e., without reference waypoints,

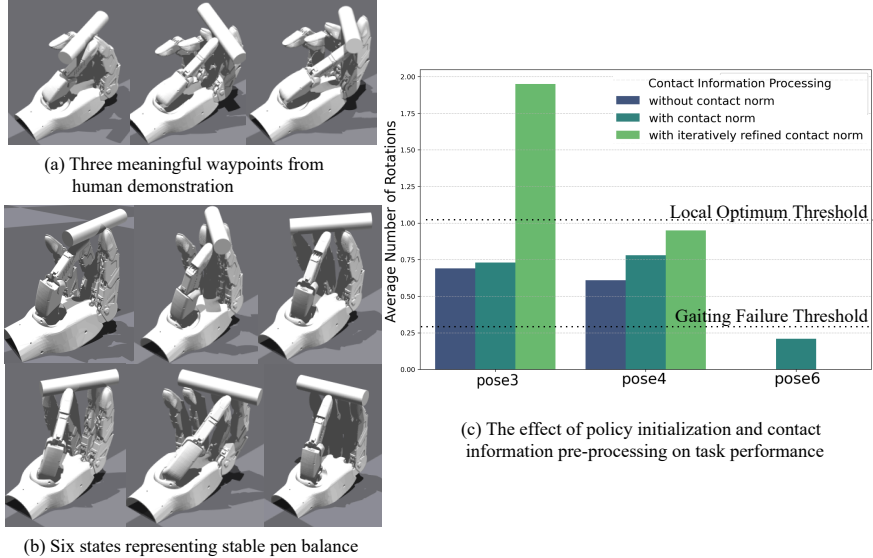


Fig. 3: Comparison of initialization strategies and their impact on performance. (a) Dynamic waypoints derived from a human trajectory. (b) Static states representing stable balancing poses. (c) Performance results show that trajectory-derived initialization (pose3) combined with iteratively refined contact normalization is crucial for success, significantly outperforming initialization from stable static states (pose6).

consistently failed to produce any successful finger gaiting behaviors. This outcome highlights the critical role of guided initialization in mastering this complex task with a high-DoF hand and aligns with prior research demonstrating the benefits of demonstrations in RL [7].

Impact of Reference Waypoint Set: The choice of reference waypoints had a marked effect on performance, as detailed in Figure 3. The initialization strategy using three waypoints from a human trajectory (Fig. 3(a)), when combined with iteratively refined normalization, yielded the highest average of 1.95 rotations. In contrast, the strategy of initializing from six stable balancing poses (Fig. 3(b)) proved significantly less effective for learning the gaiting motion, achieving a maximum of only 0.21 average rotations. While these static poses can be suitable for simpler balancing tasks on other hand morphologies, they were less effective here.

These results indicate that a concise set of waypoints depicting critical transitional stages of the desired motion is more effective for guiding exploration than a larger set of waypoints or a set of static poses. As shown in Figure 4, this superior guidance enables the discovery of policies that can sustain the motion, leading to significantly longer episode lengths and higher cumulative rewards.

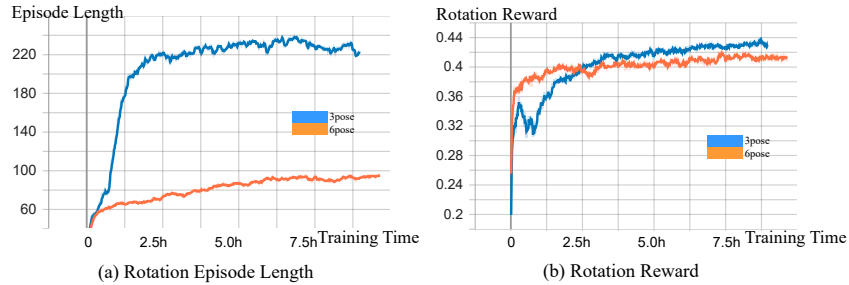


Fig. 4: Effect of initialization strategy on learning. (a) Policies initialized with dynamic waypoints (3-pose) achieve significantly longer episode durations than those initialized with static poses (6-pose). (b) Although both configurations learn the basic rotation task, as indicated by their comparable rotation rewards, only the 3-pose policy acquires a sustainable finger-gaiting strategy, enabling continuous motion and higher cumulative rewards.

4.2 Validation of Contact Force Processing

The use of detailed 3D net force vectors as privileged information, coupled with an effective normalization strategy, was crucial for learning the task. An ablation study in which these force vectors were replaced with binary contact data resulted in a significant performance decrease, underscoring that continuous net force data are more informative than simple binary signals for this complex, contact-rich manipulation task.

Furthermore, we systematically evaluated three contact force normalization methods, with results shown in Figure 3(c). For the best-performing 3-pose initialization, the baseline without normalization achieved an average of 0.69 rotations. Applying a single, fixed normalization improved performance slightly to 0.73 rotations. However, the strategy of iteratively refining the contact norm during training yielded a substantial improvement, reaching an average of 1.95 rotations. This demonstrates that adapting the normalization scheme to the policy’s evolving force-application behavior provides a more stable and effective learning signal.

4.3 Comprehensive Performance in Pen-Spinning

The optimal configuration, which paired three human-derived reference waypoints with an iteratively refined contact norm, successfully learned a stable pen-spinning policy that averaged 1.95 rotations (Fig. 3(c)). This consistent, periodic motion was a direct result of the acquired finger gaiting capabilities. As depicted in Figure 5, the learned policies demonstrated emergent and complex finger gaiting, characterized by:

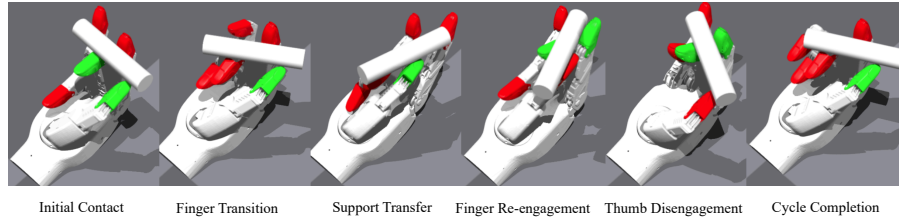


Fig. 5: The emergent finger gaiting cycle from the learned policy. The sequence shows a full rotation, featuring sophisticated, human-like coordination, such as alternating support, smooth handoffs, and anticipatory repositioning of the fingers. The states of the contact forces at the five observed fingertips are indicated by the color of their respective links. Green links correspond to forces exceeding current threshold, whereas red links represent forces below it.

1. **Multi-Contact Coordination:** The hand utilized finger pads, sides, and proximal segments, a clear departure from simpler fingertip-balancing strategies.
2. **Anticipatory Movements:** Fingers predictively repositioned themselves to prepare for subsequent contacts in the gaiting sequence.
3. **Dynamic Force Modulation:** The policy demonstrated smooth and contextually appropriate adjustments in contact forces to support, propel, and guide the pen.
4. **Adaptive Stability:** The hand maintained robust control, making real-time corrections to counteract minor disturbances.

These sophisticated behaviors starkly contrast with the static balancing common on less anthropomorphic hands [7], confirming that a suitable hand morphology paired with targeted learning strategies can foster human-like manipulation skills.

5 Discussion

5.1 Summary of Key Findings and Advantages

This research demonstrates that a high-DoF anthropomorphic hand can learn dynamic, human-like finger gaiting through a tailored RL framework. Our primary contributions include a waypoint-guided initialization strategy to overcome high-dimensional exploration challenges and the use of privileged 3D net force information to master nuanced contact dynamics. The resulting emergent gaiting, achieved with high training efficiency within 1.5 hours, validates that combining appropriate hardware morphology with targeted learning techniques can produce complex skills that surpass simpler balancing strategies. This work opens avenues for other challenging non-prehensile tasks and suggests that acquiring such skills is feasible without prohibitive computational costs.

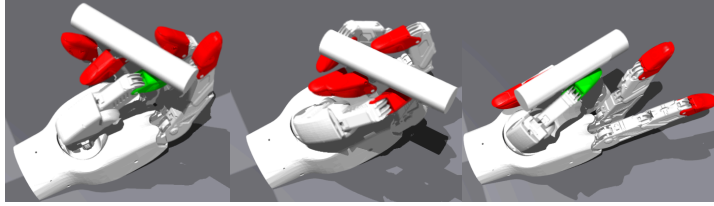


Fig. 6: An example of a policy failure state. The hand becomes trapped in a local optimum, from which it cannot continue the rotation.

Furthermore, transferring learned behaviors from low-DoF to high-DoF systems is a significant challenge in robotics. This paper offers a practical case study in this domain. We show that a reward structure from a less complex hand can be successfully migrated to a high-DoF hand by adapting physically-grounded penalties, such as joint torques, to the new embodiment. This highlights a viable method for leveraging findings from simpler systems to unlock sophisticated skills on more capable hardware, ensuring stable and efficient learning across different platforms.

5.2 Future Work

Future research will pursue several key directions. A primary goal is to avoid local optima, as illustrated in Fig.6, by developing more robust, task-agnostic reward functions and integrating richer tactile sensing modalities.

The ultimate validation of our approach hinges on transferring the learned policies to the physical world, which is the most critical future direction for this research. Achieving a successful transfer will require significant enhancements to the gaiting skill’s robustness, including enabling higher speeds, recovery from perturbations, and generalization to diverse objects. To manage the immense complexity of this task on a physical high-DoF hand, it will also be crucial to explore advanced hierarchical or bio-inspired control architectures. While the efficient demonstration of learned finger gaiting in simulation already underscores the potential for developing robots with human-like dexterity, dedicated research into this sim-to-real transfer is essential for fully realizing that potential.

6 Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grant 52475001. The authors gratefully acknowledge the Beijing Super Cloud Computing Center for providing computational resources and Linkerbot Co., Ltd. for supplying the URDF model of the dexterous hand.

References

1. Chen Tang, Ben Abbatematteo, Jiaheng Hu, Rohan Chandra, Roberto Martín-Martín, and Peter Stone. Deep reinforcement learning for robotics: A survey of real-world successes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 28694–28698, 2025.
2. OpenAI: Marcin Andrychowicz, Bowen Baker, Maciek Chociej, Rafal Jozefowicz, Bob McGrew, Jakub Pachocki, Arthur Petron, Matthias Plappert, Glenn Powell, Alex Ray, et al. Learning dexterous in-hand manipulation. *The International Journal of Robotics Research*, 39(1):3–20, 2020.
3. Jialiang Zhang, Haoran Liu, Danshi Li, XinQiang Yu, Haoran Geng, Yufei Ding, Jiayi Chen, and He Wang. Dexgraspnet 2.0: Learning generative dexterous grasping in large-scale synthetic cluttered scenes. In *8th Annual Conference on Robot Learning*, 2024.
4. Dexin Wang, Chunsheng Liu, Faliang Chang, Hengqiang Huan, and Kun Cheng. Multi-stage reinforcement learning for non-prehensile manipulation. *IEEE Robotics and Automation Letters*, 2024.
5. Xianyi Cheng, Sarvesh Patil, Zeynep Temel, Oliver Kroemer, and Matthew T Mason. Enhancing dexterity in robotic manipulation via hierarchical contact exploration. *IEEE Robotics and Automation Letters*, 9(1):390–397, 2023.
6. A. Bicchi. Hands for dexterous manipulation and robust grasping: a difficult road toward simplicity. *IEEE Transactions on Robotics and Automation*, 16(6):652–662, 2000.
7. Jun Wang, Ying Yuan, Haichuan Che, Haozhi Qi, Yi Ma, Jitendra Malik, and Xiaolong Wang. Lessons from learning to spin "pens". *arXiv preprint arXiv:2407.18902*, 2024.
8. Yuanpei Chen, Tianhao Wu, Shengjie Wang, Xidong Feng, Jiechuang Jiang, Stephen Marcus McAleer, Yiran Geng, Hao Dong, Zongqing Lu, Song-Chun Zhu, and Yaodong Yang. Towards human-level bimanual dexterous manipulation with reinforcement learning, 2022.
9. Ashvin Nair, Bob McGrew, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Overcoming exploration in reinforcement learning with demonstrations, 2018.
10. Shuo Cheng and Danfei Xu. League: Guided skill learning and abstraction for long-horizon manipulation. *IEEE Robotics and Automation Letters*, 8(10):6451–6458, 2023.
11. Haozhi Qi, Brent Yi, Sudharshan Suresh, Mike Lambeta, Yi Ma, Roberto Calandra, and Jitendra Malik. General in-hand object rotation with vision and touch. In *Conference on Robot Learning*, pages 2549–2564. PMLR, 2023.
12. Soroush Nasiriany, Huihan Liu, and Yuke Zhu. Augmenting reinforcement learning with behavior primitives for diverse manipulation tasks. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 7477–7484. IEEE, 2022.
13. Tao Chen, Jie Xu, and Pulkit Agrawal. A system for general in-hand object reorientation. In *Conference on Robot Learning*, pages 297–307. PMLR, 2022.
14. Tao Chen, Megha Tippur, Siyang Wu, Vikash Kumar, Edward Adelson, and Pulkit Agrawal. Visual dexterity: In-hand reorientation of novel and complex object shapes. *Science Robotics*, 8(84):eadc9244, 2023.
15. John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.